

Education

2024-PRESENT	Ph.D. in Computer Sciences, UC Berkeley Advised by Prof. Laurent El Ghaoui. College of Engineering Ph.D. Commencement Student Speaker. Affiliated with Berkeley AI Research (BAIR) Lab, NSF AI Institute for Advances in Optimization (AI4OPT), and Pacific Earthquake Engineering Research (PEER) Center.
2020	M.S. in Information Management and Systems, UC Berkeley Affiliated with Center for Technology, Society & Policy, and The Complexity Science for Nuclear Security Group
2018	M.S. in Computer Science, Georgia Institute of Technology Transferred to UC Berkeley.
2016	Visiting Student, Dept. of Sociology, UC Berkeley
2016	B.B.A. in International Business, National Taiwan University Graduate with Distinction. President at Modern Dance Club.

Employment

PRESENT	Research Engineer, Google DeepMind
FALL 2023	Research Intern, Google Part of Google DeepMind and Play Team. Research on large language models (LLMs) for recommendation systems.
SUMMER 2023	Machine Learning Engineering Intern, Apple Part of the AI/ML Siri Understanding team. Research on cost-effective data-enteric automatic speech recognition.
2022-2023	Research Assistant, College of Engineering & Computer Science, VinUniversity Research on the theories, computational algorithms, and applications of implicit models.
SUMMER 2022	Applied Scientist Intern, Amazon Alexa AI Part of the Creative AI team. Offensive content detection and mitigation in narrative story generation for kids.
SUMMER 2021	Applied Scientist Intern, Amazon Alexa AI Part of Natural Language Generation (NLG) team. Affect-aware controllable response generation in open-domain dialog for understanding and identifying user emotions.
SUMMER 2020	Applied Scientist Intern, Amazon Alexa AI Part of NLG team. Stylistic control for schema-guided natural language generation in task-oriented dialog systems.
SUMMER 2020	Artificial Intelligence Researcher, SumUp Analytics Efficient unsupervised topic modeling and document summarization for large-scale corpus.
SUMMER 2019	Data Science Intern, LinkedIn Predictive modeling for business-to-business (B2B) client acquisition.
2018-2019	Graduate Student Researcher, Dept. of Nuclear Engineering, UC Berkeley Advised by Dr. Bethany L. Goldblum. Nuclear activities and proliferation detection with physical sensing data.
SUMMER 2018	Research Intern, Institute for Creative Technology, University of Southern California Modeling user learning records. Developed backend services for data conversion in intelligent tutoring systems.
2017-2018	Research Assistant, College of EECS, National Taiwan University Advised by Prof. Shou-De Lin. CNN and GNN for 3D spatial and geological data for oil reservoirs. Cross-domain recommender system with common-sense knowledge. Maximum flow algorithms for retail construction plan with optimal customer satisfaction.
2016-2017	Investment Associate (2016-2017), Analyst (2015-2016), Cherubic Ventures Deal sourcing, due diligence, market evaluation, and investments assessment in tech and AI sector for seed to series-B companies in the U.S. and East Asia. Portfolio management and investor relations.
2014-2015	Research Assistant, Institute of Economics, Academia Sinica. Advised by Prof. Stacey H. Chen. Socio-economic influence on health insurance decision making.

Publications

BOOK & BOOK CHAPTERS

1. Laurent El Ghaoui, Alicia Y. Tsai, and Giuseppe C. Calafiore. *“Linear Algebra and Applications”*. VinUniversity, 2023
2. Giuseppe C. Calafiore, Laurent El Ghaoui, Giulia Fracastoro, and Alicia Tsai. *“Financial Data Science”*. Cambridge University Press & Assessment, 2024

PEER-REVIEWED

1. Alicia Y. Tsai, Shereen Oraby, Anjali Narayan-Chen, Alessandra Cervone, Nanyun Peng, Spandana Gella, Tagyoung Chung, and Jing Huang. *“Taxonomy, Dataset, and Evaluation in Ensuring Child-Safe Content for Story Generation”*. In *Empirical Methods in Natural Language Processing (EMNLP) Workshop on NLP for Positive Impact*, 2024
2. Juliette Decugis, Max Franklin Emerling, Ashwin Ganesh, Alicia Y. Tsai, and Laurent El Ghaoui. *“The Extrapolation Power of Implicit Models”*. In *International Joint Conference on Artificial Intelligence (IJCAI) Workshop on Explainable Artificial Intelligence (XAI)*, 2024
3. Alicia Y. Tsai, Wenzhi Gao, and Laurent El Ghaoui. *“Constrained Implicit Learning Framework for Neural Network Sparsification”*. In *Asian Conference on Machine Learning (ACML)*, 2024
4. Alicia Y. Tsai, Adam Kraft, Long Jin, Chenwei Cai, Anahita Hosseini, Taibai Xu, Zemin Zhang, Lichan Hong, Ed H. Chi, and Xinyang Yi. *“Leveraging LLM Reasoning Enhances Personalized Recommender Systems”*. In *Findings of the Association for Computational Linguistics (ACL)*, 2024
5. Juliette Decugis, Max Emerling, Ashwin Ganesh, Alicia Y. Tsai, and Laurent El Ghaoui. *“On the Abilities of Mathematical Extrapolation with Implicit Models”*. In *Neural Information Processing Systems (NeurIPS) Workshop on Distribution Shifts: Connecting Methods and Applications*, 2022
6. Alicia Y. Tsai, Shereen Oraby, Vittorio Perera, Jiun-Yu Kao, Yuheng Du, Anjali Narayan-Chen, Tagyoung Chung, and Dilek Hakkani-Tur. *“Style Control for Schema-Guided Natural Language Generation”*. In *Empirical Methods in Natural Language Processing (EMNLP) Workshop on Natural Language Processing for Conversational AI*, 2021
7. Laurent El Ghaoui, Fangda Gu, Bertrand Travacca, Armin Askari, and Alicia Y. Tsai. *“Implicit Deep Learning”*. In *SIAM Journal on Mathematics of Data Science*, 2021
8. Alicia Y. Tsai, Selim Günay, Minjune Hwang, Pengyuan Zhai, Chenglong Li, Laurent El Ghaoui, and Khalid M. Mosalam. *“Text Analytics for Resilience-Enabled Extreme Events Reconnaissance”*. In *Neural Information Processing Systems (NeurIPS) Workshop on Artificial Intelligence for Humanitarian Assistance and Disaster Response*, 2020
9. Alicia Y. Tsai and Laurent El Ghaoui. *“Sparse Optimization for Unsupervised Extractive Summarization of Long Documents with the Frank-Wolfe Algorithm”*. In *Empirical Methods in Natural Language Processing (EMNLP) Workshop on Simple and Efficient Natural Language Processing*, 2020
10. Geoffrey Negiar, Gideon Dresdner, Alicia Yi-Ting Tsai, Laurent El Ghaoui, Francesco Locatello, and Fabian Pedregosa. *“Stochastic Frank-Wolfe for Constrained Finite-Sum Minimization”*. In *International Conference on Machine Learning (ICML)*, 2020
11. Khalid M. Mosalam, Sifat Muin, Selim Günay, Alicia Y. Tsai, Minjune Hwang, Chenglong Li, Jiang Lian, and Laurent El Ghaoui. *“Building Resilience Through Structural Health Monitoring and Reconnaissance”*. In *World Conference on Earthquake Engineering (WCEE)*, 2020
12. Yi-Ting Tsai, Min-Chu Yang, and Han-Yu Chen. *“Adversarial Attack on Sentiment Classification”*. In *Association for Computational Linguistics (ACL) Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP*, 2019
13. Yi-Ting Tsai, Chih-Shiang Wuy, Hsiang-Ling Hsuy, Tenniel Liuy, Pei-Lin Cheny, Wen-Hao Chen, and Keng-Te Liao. *“A Cross-Domain Recommender System Based on Common-Sense Knowledge Bases”*. In *Conference on Technologies and Applications of Artificial Intelligence (TAAI)*, 2017

REPORTS

1. Eduardo Miranda, Andres Acosta Vera, Luis Aponte, Jorge Archbold, Maria Cortes, Ao Du, Selim Gunay, Wael Hassan, Pablo Heresi, Ana Lamela, Armando Messina, Sebastian Miranda, Jamie Padgett, Alan Poulos, Giulia Scagliotti, Alicia Tsai, Tracy Kijewski-Correa, Ian Robertson, Khalid Mosalam, David Prevatt, and David Roueche. **“StEER - 07 Jan. 2020 Puerto Rico Mw6.4 Earthquake: Preliminary Virtual Reconnaissance Report”**. Structural Extreme Events Reconnaissance Network, 2020
2. Selim Gunay, Jorge Archbold, Fan Hu, Alicia Tsai, Khalid Mosalam, Tracy Kijewski-Correa, Ian Robertson, David Prevatt, and David Roueche. **“StEER - 15 December 2019 Earthquake in the Philippines: Event Briefing”**. Structural Extreme Events Reconnaissance Network, 2019
3. Alicia Tsai, Fan Hu, Selim Gunay, Khalid Mosalam, Tracy Kijewski-Correa, Ian Robertson, David Prevatt, and David Roueche. **“StEER - 26 November, 2019 Albania, Mw 6.4 Earthquake: Event Briefing”**. Structural Extreme Events Reconnaissance Network, 2019

PATENTS

1. Jared Zhao, Bethany L. Goldblum, Christopher Stewart, Alicia Yi-Ting Tsai, Shruthi Chockkalingam, and Pedro Vicente Valdez. **“MIMOSAS: Multimodal Input Model Output Security Analysis Suite”**. Computer software. University of California Berkeley, 2019

Presentations & Selected Talks

MAY 2024	Deep Implicit Time Series Modeling for Earthquake Phase Picking on Edge Devices , 2024 Seismological Society of America Annual Meeting
JAN 2023	State-driven Implicit Modeling , Third Workshop on Seeking Low-Dimensionality in Deep Neural Networks
DEC 2022	Implicit Deep Learning , VinAI Research Seminar
NOV 2022	Implicit Deep Learning , College of Engineering & Computer Science, VinUniversity
APR 2022	Big Data Ethics and Responsible AI , Invited Lecture for The Last Humans & Earth: Identifying Crisis in Sustainability, Taipei Medical University
MAY 2021	Returning to Foster Care: Revisited with Computational Text Analysis , UC Berkeley D-Lab Computational Text Analysis Working Group
OCT 2020	Sparse Optimization for Unsupervised Extractive Summarization of Long Documents with the Frank-Wolfe Algorithm , West Coast NLP (WeCNLP) Summit
OCT 2020	Robustness Analysis of Deep Learning via Implicit Models , Bay Area Machine Learning (BayLearn) Symposium
JAN 2020	Natural Language Processing and Text Analytics for Earthquake Reconnaissance , The Pacific Earthquake Engineering Research (PEER) Annual Meeting
JUL 2019	Adversarial Attack on Sentiment Classification , Widening Natural Language Processing (WiNLP) Workshop, ACL

Teaching & Tutoring

SPRING 2023	Linear Algebra , Teaching Assistant, VinUniversity
FALL 2022	Linear Algebra , Teaching Assistant, VinUniversity
2022-2023	High School Science Fair Project , Private Tutor
FALL 2022	Financial Data Science , Graduate Student Instructor, UC Berkeley
SPRING 2021	Optimization Models in Engineering , Graduate Student Instructor, UC Berkeley
SPRING 2020	Natural Language Processing , Graduate Student Instructor, UC Berkeley
FALL 2019	Introduction to Machine Learning and Data Analytics , Graduate Student Instructor, UC Berkeley
SPRING 2018	Operation Research , Teaching Assistant, National Taiwan University
SPRING 2018	Python for Machine Learning , Co-Instructor, Coding and Co-working Club, National Taiwan University
FALL 2017	Programming for Business Computing , Teaching Assistant, National Taiwan University
FALL 2017	Flask Web Development , Co-Instructor, PyLadies Taiwan

FALL 2017	Python for Data Analysis , Co-Instructor, Coding and Co-working Club, National Taiwan University
SPRING 2015	Statistical Data Analysis for Business and Management , Teaching Assistant, National Taiwan University
2012–2018	K-12 Math, Science, and English, College English , Private Tutor

Awards & Honors

SCHOLARSHIPS & FELLOWSHIPS

2024	Thesis Research Fellowship , Kapor Foundation
2024	The Rising Star Women in Engineering , Asia Dean's Forum
2024	Awards for Undergraduate Researcher Mentoring , EECS Department, UC Berkeley
2024	Research Fellow , Bain Venture Capital
2023	300 Hours Teacher Training BIPOC Scholarship , Down Under School of Yoga
2023	STARS AI Scholars , The STARS Computing Corps, supported by the National Science Foundation (NSF)
2023	Finalist for Rising Star in Machine Learning , University of Maryland Center for Machine Learning
2023	200 Hours Teacher Training BIPOC Scholarship , Nomad Yoga
2023	Awards for Undergraduate Researcher Mentoring , EECS Department, UC Berkeley
2021	The Ministry of Education Study Abroad Scholarship , Taiwan
2021	Generation Google Scholarship (North America) , Google
2020	Fellow , Center for Technology, Society and Policy, UC Berkeley
2020	Scholar , Computing Research Association-Widening Participants, Grad Cohort for Underrepresented Minorities and Persons with Disabilities (URMD)
2018, 2019	Scholar , Computing Research Association-Women, Grad Cohort for Women
2018	NSF Scholar , ACM Richard Tapia Celebration of Diversity in Computing
2018	Scholar , Grace Hopper Celebration
2018	40 Under 40 Women in Tech Award , Girls in Tech Taiwan
2017	Special Project and Philanthropy Award , Cathy Charity Foundation
2015	Scholarship for Future Scholars in Humanities and Social Science , Ministry of Education, Taiwan
2015	Dr. Chien, Shih-Liang Memorial Scholarship , National Taiwan University
2013, 2014	Dean's List , National Taiwan University
2014	Dr. Yu, Kuo-Hwa Memorial Scholarship , National Taiwan University
2013	Samsung Scholarship , National Taiwan University
2012	Academician Chu, Cyrus C. Y. and Prof. Chen, Tain-Jy Memorial Scholarship , College of Management, National Taiwan University

GRANTS

2024	Research Grant , Cohere For AI
2023	Travel Grant , ACM Richard Tapia Celebration of Diversity in Computing
2023	Travel Grant , The Workshop on Seeking Low-Dimensionality in Deep Neural Networks
2022	Research Grant , Hearts to Humanity Eternal (H2H8) Association
2022	Travel Grant , Computing Research Association-Grad Cohort Workshop for Inclusion, Diversity, Equity, Accessibility, and Leadership Skills (IDEALS)
2022	Travel Grant , The Workshop on Robustness and Resilience in Stochastic Optimization and Statistical Learning: Mathematical Foundations, The Ettore Majorana Foundation and Centre for Scientific Culture
2021	Google Cloud Research Credits
2020	STEM First-Year Initiative Professional Development Grant , UC Berkeley
2020	Travel Grant , EMNLP Workshop on Simple and Efficient Natural Language Processing
2019	Travel Grant , ACL Workshop on BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP
2019	Travel Grant , ACL Workshop on Widening Natural Language Processing
2015	Undergraduate Research Grant , Ministry of Science and Technology, Taiwan

Professional Services & Activities

REVIEWER	Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024), Machine Learning Safety Workshop, NeurIPS (2022), The American Control Conference (2022), Africa Natural Language Processing Workshop, EACL (2021), Workshop on Uncertainty and Robustness in Deep Learning, ICML (2020)
AREA CHAIR	Women in Machine Learning Workshop, NeurIPS (2019, 2020)

LEADERSHIP ROLES

2022-2023	Student Leadership Council , Artificial Intelligence Institute for Advances in Optimization (AI4OPT)
2021-2022	Advisory Council , Global Women in Data Science (WiDS) Conference
2020-PRESENT	Founding Board Member , Taiwan Data Science Association
2020-2023	Board Secretary , Women in Machine Learning (WiML)
2020-2022	Founding Member , Taiwanese in Data Science
2019-2023	Co-Organizer , Women in Data Science (WiDS) Conference at Berkeley , UC Berkeley
2020-2021	Co-Lead , Global Teaching Fellowship Program , Delta Analytics
2020-2021	Co-Chair , Widening Natural Language Processing (WiNLP) Workshop , ACL
2017-2019	Founder and Organizer , Women in Data Science (WiDS) Conference at Taipei
2017-2019	Engineering Partner , Dorm Room Fund
2017	Program Committee and Sponsorship Team , Python Conference (PyCon) Taiwan

DEPARTMENTAL SERVICES

SPRING 2024	Ph.D. Commencement Ceremony Student Speaker , College of Engineering, UC Berkeley
FALL 2023	Textbook Cover Design Judging Panel , College of Engineering & Computer Science, VinUniversity
SPRING 2022	CS Faculty Interview Student Committee , EECS Department, UC Berkeley
SPRING 2022	Graduate Application Review Committee, Student Reader , EECS Department, UC Berkeley
SPRING 2022	Visit Day Organizer , Control, Intelligent Systems and Robotics Program, EECS Department, UC Berkeley

OUTREACH & MENTORSHIP

SUMMER 2023	Research Mentor , Transfer-to-Excellence Summer Research Program , UC Berkeley
2022-2023	Ambassador , STEM First Year Initiatives , UC Berkeley
2022-2024	Graduate Mentor , Getting into Graduate School (GiGS) Mentorship Program , UC Berkeley
2021	Graduate Mentor , Equal Access to Application Assistance Program , EECS Department, UC Berkeley
2020	Graduate Mentor , Berkeley Graduate Application Mentoring Program, UC Berkeley
2020, 2022	Graduate Application Mentoring Program , Project Tyra – Taiwanese Young Researcher Association
2020, 2021	Mentoring Program , Taiwan Data Science Association
2019-2021	Bay Area Teaching Fellow , Delta Analytics

PRESS, INTERVIEWS, & PANEL

2024	College of Engineering Doctoral Commencement , UC Berkeley
2024	Navigating Google and Beyond , Carpe Datum Podcast
2023	Graduate School Applications Sharing Panel , Synergy Consultant Group, Donz GMAT, & Lily GRE
2021	Data Science Career Panel , National Taiwan University Data Analytics Club
2021	Women in Data Science Berkeley – Featuring Alicia Tsai , UC Berkeley Cloud Computing Meetup
2021	Don't Miss the Women in Data Science Conference: March 8–12 , University of California IT Blog
2020	Female Engineers Working in Silicon Valley , YouTube Interviewed in Mandarin Chinese
2018	Inclusion for Data Science in Taiwan , Bay Area and Taiwan Data Science Meetup
2018	Women of Data Science – Alicia Tsai Launches First WiDS Conference in Taiwan , Western Digital
2018	女孩領軍-WiDS 女性資料科學論壇在台北 , 網氏 / 閩市女性電子報